

Wi-SUN DCU

NLG6770-IN



Introduction

Wi-SUN DCU (NLG6770-IN) provides secured interconnection of Wi-SUN field devices with WAN communication networks. Its capabilities include Wi-SUN FAN 1.0 Support, remote device management, communication management, data storage, and interface abstraction. Having choice of multiple WAN interfaces and mounting options makes it suitable for different deployment strategies and locations. Ideal for small to medium mesh network supporting up to 200 nodes on single network for various IoT applications.

Applications

- Smart Utility Metering (Electricity, Gas, Water)
- Industrial Automation & Predictive Maintenance
- Agriculture & Environmental Monitoring
- Smart Street Lighting
- Defence, Military Applications
- Smart Cities

Features

- Wide Input Supply Range (upto 528VAC) with Integrated over-voltage protection
- Support upto 200 nodes on single network (*based on field scenario this will change)
- Offline data storage of 32 GB
- Tamper detection
- External fiber glass antenna with 5 dBi gain
- 10kV surge protection
- Support DCU with battery backup 4 hrs
- LTE, Ethernet, Wi-Fi (Optional)

Processor	
ARM® Cortex®-A7 Microprocessor IC i.MX6 1 Core, 32-Bit 792MHz 289-MAPBGA	
Operating System	
Yocto Based Linux System	
Storage	
256MB NAND Flash, 512MB DDR3L RAM, 32 GB SD CARD	
Wi-SUN	
Frequency	IN 865-868 MHz (Other Country Options Available. Refer Product Selector List)
Conductive Power	+26 dBm (Configurable Based on region)
Receiver Sensitivity	-109 dBm @50 kbps 2FSK Signal -107 dBm @100 kbps 2FSK Signal
Antenna	External 5dBi Fiber Glass Antenna
Cellular	
Cellular bands	LTE FDD - B1/B3/B5#/B7/B8/ B20/B28# GSM - GSM/GPRS/EDGE 900/1800 MHz
SIM card	Nano SIM Tray
Antenna	Inside Enclosure
Wi-Fi (Optional-Refer Product Selector Guide)	
Band	Single Spatial Stream in 2.4 GHz ISM Band
Standards	802.11 b/g/n, 802.3, 802.3u 802.11e-compatible bursting and I standards
Antenna	Inside Enclosure
Ethernet	
1 x IEEE 802.3 10/100M	
USB	
TYPE C for Serial Debug Communication	
Power Supply	
Power Supply	85 ~ 528VAC
Power Backup	4 hrs Battery Backup After Power Failure
Power Consumption	Average 10W (Approx)
Mechanical Specification	
Size	250 x 210 x 92 mm
Mounting Type	Pole / Wall Mount
Enclosure	Polycarbonate - UV Stabilized, UL94 V0 Rated, IP65 Self Certified, Fire retardant rated case
Environmental Specification	
Operating Temperature	-10°C to 60°C
Humidity	Relative Humidity 15% to 93% Non Condensing
Pre-Compliance Certification	
- ESD (IEC 61000-4-2) - EFT (IEC 61000-4-4) - Surge (IEC 61000-4-5) - Radiated Emission (CISPR22) - Test of immunity to electromagnetic HF field (IEC61000-4-3)	

For more information contact at <https://www.nebulae.io> or sales@nebulae.io

System Level Solutions

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Model Structure

Model Example

NLG6770 – IN

Based SLS DCU name

NLG6 – SLS DCU code

Feature Selector to Connect Cloud/Server

5 – GSM, Ethernet

7 – GSM, Wi-Fi, and Ethernet

Radio

70 – Wi-SUN

Wi-SUN Region

IN (IN865)--865–867 MHz--India, Pakistan, Niger (India band)

US (US915)--902–928 MHz--USA, Canada, Mexico (North America)

EU (EU868)--863–870 MHz--Europe + many EMEA countries (e.g., UK, France, RSA, Saudi Arabia, UAE, Turkey, Malta, Switzerland)

AU (AU915)--915–928 MHz--Australia, New Zealand, Argentina, Chile, Brazil [often AU915 band allowed]

RU (RU864)--864–870 MHz--Russia regulatory band

Ordering Code: PL1WSNSOM0100

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